3.5.2 INSTRUMENT CLUSTER SELF TEST

The Instrument Cluster is capable of performing a diagnostic self test. The self test is actuated by depressing and holding the trip reset button while turning the ignition from the off to the on position. The self test can also be initiated using the DRBIII®. The self test will terminate if the tachometer is greater than 300 RPM, the ignition is turned off, or the test is complete. For diagnostic procedures, refer to the TJ Body Diagnostic Procedures Manual.

3.6 VEHICLE COMMUNICATION

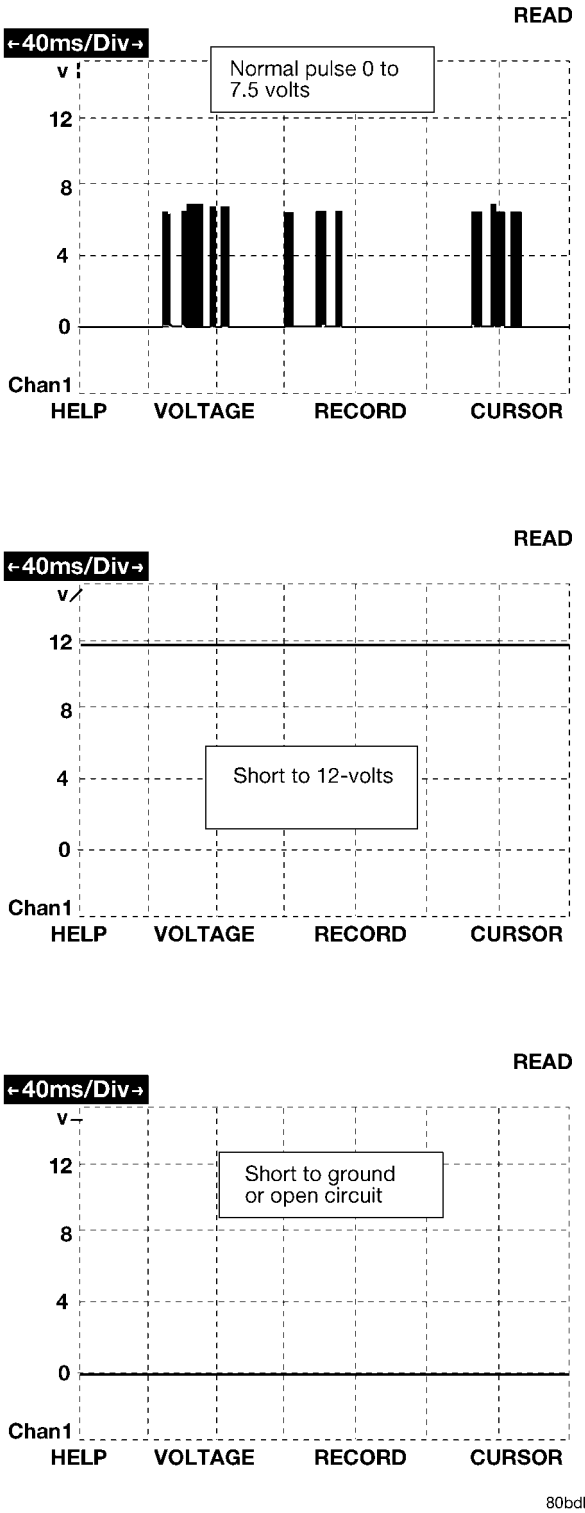
The Programmable Communication Interface or PCI Bus is a single wire multiplexed network capable of supporting binary encoded messages shared between multiple modules. The PCI bus circuit is identified as D25 and is violet with a yellow tracer. The modules are wired in parallel. Connections are made in the harness using splices. The following modules are used on 2005 Jeep® Wrangler (TJ).

- Airbag Control Module
- Mechanical Instrument Cluster
- Radio
- Powertrain Control Module (NGC)
- Sentry Key Immobilizer Module
- Controller Antilock Brake (Not on the PCI Bus. Communication with the scan tool is established thru the SCI Transmit circuit).

Each module provides its own bias and termination in order to transmit and receive messages. The bus voltage is at zero volts when no modules are transmitting and is pulled up to about seven and a half volts when modules are transmitting.

The bus messages are transmitted at a rate averaging 10800 bits per second. Since there is only voltage present when the modules transmit and the message length is only about 500 milliseconds, it is ineffective to try and measure the bus activity with a conventional voltmeter. The preferred method is to use the DRBIII® lab scope. The 12v square wave selection on the 20-volt scale provides a good view of the bus activity. Voltage on the bus should pulse between zero and about seven and a half volts. Refer to the following figure for some typical displays.

The PCI Bus Failure modes are broken down into two categories. Complete PCI Bus Communication Failure and individual module no response. Causes of a complete PCI Bus Communication Failure include a short to ground or battery on the PCI circuit. Individual module no response can be caused by an open circuit at the module, or an open battery or ground circuit to the affected module.



Symptoms of a complete PCI Bus Communication Failure would include but are not limited to:

- All gauges on the MIC stay at zero
- All telltales on MIC illuminate
- MIC backlighting at full intensity
- No response received from any module on the PCI bus (except PCM)
- No start (if equipped with Sentry Key Immobilizer)